Sahaj Bhadja

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Summary

Full-stack engineer with 2+ years building **platform services**, **REST APIs**, and **production-grade applications** across AWS, Docker, and Kubernetes, with experience owning full-lifecycle feature development from prototype to deployment.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife — **Los Angeles, CA**
Software Engineer — *September 2025 – Present*

- Built **platform services and REST APIs** for 500+ users, reducing request latency from **1.2s to 350ms** by optimizing execution paths, tuning PostgreSQL queries, and improving responsiveness for mission-critical analytics workflows.
- Implemented **automated validation and CI/CD pipelines** for 5,000+ monthly jobs, reducing downstream failures and improving production reliability through earlier detection of regressions and data-quality drift.

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*

- Developed **full-stack features and REST APIs** handling 10,000+ daily requests, owning full-lifecycle development from prototype to production while partnering with cross-functional teams in a fast-paced startup environment.
- Built **prototypes for AI-assisted workflows** with multimodal models, validating technical constraints and iterating from proof-of-concept to working product features serving 150+ users in production.

USC AD Labs — **Los Angeles, CA**
ML Research Assistant — *January 2025 – May 2025*

- Built **large-scale experimentation pipelines** across 100,000+ records, understanding tradeoffs in competing approaches and articulating technical constraints to senior researchers while contributing to implementation progress.
- Developed **prototypes to test architectural concepts**, validating design decisions through experimentation and evolving successful prototypes into production-ready model training infrastructure.

Aasma Technology Solutions — **India**
Software Developer Engineer - (SecureSync) — *August 2023 – July 2024*

- Engineered **API microservices and ETL pipelines** using **AWS Glue and S3**, improving response time by **30%** while following software engineering best practices and contributing to feature development and bug fixes.
- Maintained **20+ automated test suites** using **Jenkins CI/CD**, PyTest, and Mockito, increasing pipeline reliability by **68%** and actively mentoring team members on testing practices and engineering culture.

Aasma Technology Solutions — **India**
Full Stack Developer Intern - (VibrantChem) — *May 2023 – August 2023*

- Developed **full-stack platform with microservices architecture** using MERN stack, WebSockets, and RESTful APIs, syncing 2M+ records with **33% fewer conflicts** and real-time data synchronization.
- Automated **Docker-based CI/CD workflows** with Jenkins, cutting release time by **50%** and improving deployment consistency across development, staging, and production environments.

Skills

Languages: **Java**, **Python**, **TypeScript**, JavaScript, C++, **SQL**, Bash
Cloud & Infrastructure: **AWS** (EC2, S3, ECS, CodePipeline, Glue), **Docker**, **Kubernetes**, Terraform basics
CI/CD & DevOps: **Jenkins**, GitHub Actions, **Git**, automated testing, deployment pipelines
Databases: **PostgreSQL**, MySQL, SQL Server, **MongoDB**, Firebase, query optimization
Frontend: React, **Next.js**, **TypeScript**, TailwindCSS, REST APIs, responsive UI

Projects

Linguistix Code | *Python, AWS CodePipeline, ECS/Fargate, Docker, CI/CD*

- Built a **containerized REST API on AWS** with CodePipeline and ECS/Fargate deployment, automated validation gates, and Docker-based workflows, achieving production-grade reliability and streamlined troubleshooting.

FinNexus Live | *Next.js, TypeScript, MongoDB, Kubernetes, WebSockets*

- Built a **scalable platform with Kubernetes deployment** supporting 50,000+ users and 1M+ transactions, with real-time WebSocket updates, MongoDB persistence, and responsive Next.js interfaces.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, MySQL, Docker, Maven*

- Built an **AI-powered SQL workspace** supporting 10+ databases (PostgreSQL, MySQL, SQLite), working across Java backend and TypeScript frontend with Docker-based development and cross-platform compatibility.